PAVITHRA MANOGARAN

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EDUCATION

Bachelors in Computer Science and Engineering 2022 - 2026
St. Joseph's College Of Engineering CGPA: 8.93

Higher Secondary Education (Class XII) 2022
Babaji Vidhyashram Percentage: 92.2%

Secondary School Education (Class X) 2020
Babaji Vidhyashram Percentage: 86.2%

EXPERIENCE

SAP Analyst Intern - Kaar Technologies Feb 2026 - Present (On-site)

- Completed training in SAP technical and functional modules and gained hands-on knowledge of the core business processes.
- Identified AI use cases of high-impact across live customer projects to demonstrate measurable business value for clients.
- Measured and reported time and cost savings from AI adoption and provided ROI insights to stakeholders.
- Collaborated with customers to strengthen engagement, health and retention.

Cloud/SRE/DevOps & GenAI Deep Dive Intern - G5Infotech May 2025 - August 2025 (Remote)

- Built a cloud-based financial tracker by using AWS services for managing services scalably and reliably.
- Integrated GenAI capabilities to analyze spending patterns of users and generate financial insights.
- Applied Site Reliability Engineering (SRE) practices like monitoring, logging and automated alerting to ensure uptime and system reliability.

Cloud Engineer Intern - HiAiDo Cloud Automation Pvt. Ltd. Sept 2024 - March 2025 (Remote)

- Built a logging pipeline using Terraform and AWS services for monitoring and analysis by collecting, storing and processing CloudTrail logs.
- Designed and implemented automated infrastructure provisioning to manage and deploy cloud resources.
- Developed a chatbot for code generation and review to enhance development workflows.

PROJECTS

AgentForge - Prompt-to-Agent Workspace:
LangGraph, MCP, FastAPI, Next.js, PostgreSQL, ChromaDB

- Developed a platform that converts natural-language prompts into agent specifications that is validated and executed through a LangGraph workflow using MCP servers.
- Implemented a semantic checker with an inline question flow to identify and resolve specification errors.
- Designed project-level isolation for chats, agents, knowledge bases and memory with an option to export developed agents as standalone Docker bundles.

VelocityLB - Adaptive Layer-7 Load Balancer:
Go, Prometheus, Grafana, Docker Compose

- Built four routing strategies: round-robin, least-connections, Power of Two Choices, and EWMA latency-aware routing with active and passive health checking and AIMD concurrency limiting.
- Benchmarked the load balancer across different routing strategies.
- Adaptive routing lowered p99 latency below 100ms while round-robin reached 503ms which showed a 5–7x performance difference using a custom load generator with Prometheus and Grafana monitoring.

Autonomous Site Reliability Engine:
Python, Kubernetes, Ollama, Qdrant, Streamlit

- Built a self-healing Kubernetes reliability system to detect, diagnose and remediate real-time system failures autonomously using a multi-agent architecture.
- Integrated Ollama LLMs with RAG using Qdrant to analyze incidents, identify root causes and generate remediation commands.
- Developed an end-to-end feedback loop with monitoring, automated recovery and verification.

GreenML - Energy-Efficient AutoML Framework:
AutoGluon, CodeCarbon, Pymoo, Flask, Streamlit, Docker

- Built an AutoML framework which optimizes machine learning models by reducing power consumption while maintaining high performance.
- Developed an optimization approach to balance accuracy metrics (F1-score, precision) using real-time CPU/GPU energy to identify efficient model configurations.
- Integrated resource monitoring, model pruning and lightweight models to make AI deployment efficient on low-power and edge devices.

Ops-Center - Enterprise War Room:
React, Node.js, Express, MongoDB, Redis, Socket.io

- Developed a real-time platform to manage incident response and corporate operations using a cloud-native architecture.
- Built a low-latency system for live incident streaming, instant task updates and team collaboration.
- Implemented WebSocket-based communication and role-based access control to manage real-time updates, tasks and decision-making during critical events.

Quota Policy Sandbox - RL-Based Resource Policy Testing:
PettingZoo, Stable-Baselines3, FastAPI, Next.js

- Built a reinforcement-learning simulator where independent agents learn adaptive and non-scripted behavior using a resource-allocation policy.
- Conducted controlled experiments by varying population size, penalties and inter-tenant visibility to determine the factors leading to fair and unfair outcomes.

PUBLICATIONS

- **An Agentic AI Framework for Dynamic Data Fusion in Flood Forecasting** — Built a multi-agent system using an LLM to combine LSTM, DeepLabv3+ and DistilBERT outputs from time-series, satellite and social media data. The system automatically adjusted the model weights based on their performance and achieved 93.5% accuracy and 0.96 AUC-ROC with a 3% improvement in static averaging.(Publication Link)
- **Agentic AI-Powered Electricity Theft Detection Using Temporal Convolution Networks and Context-Aware Reasoning** — Designed a hybrid framework using TCN to analyze smart-meter time-series data and implemented context-aware module using historical and seasonal patterns to reduce false positives and provide accurate theft-risk predictions.(Publication Link)

PATENTS

- **AI-Powered Career Guidance Aligned with NEP 2020**: A Personalized and Scalable Platform for Students — Indian Patent Application No. 202541128458, published 20 Feb 2026.
  - Mobile application which uses students' academic records, skills and certifications to suggest suitable career paths, internships and certifications.

- **CloudSecAgent**: Secure AI Agent Framework for Cloud Security Compliance — Indian Patent Application No. 202641031745, published 27 Mar 2026.
  - Agentic framework which audits cloud systems against regulatory guidelines using real-time system data rather than assumptions generated by AI.

SKILLS

AI/ML	Scikit-Learn, TensorFlow, LangChain, Agentic AI, LangGraph AWS Bedrock, Prompt Engineering, RAG Systems, Model Evaluation
Programming Languages	Python, Java, JavaScript, HTML, CSS
Frameworks & APIs	React, FastAPI, Flask, REST APIs, SAP ABAP, OData, UI5
Backend & Database	Node.js, Express, MySQL, MongoDB, ChromaDB
Cloud & DevOps	Amazon Web Services (AWS), Microsoft Azure, Docker, Kubernetes, Terraform, Jenkins, Prometheus, Grafana
Version Control & Tools	Git, GitHub, Linux, VS Code, Postman, Ollama

COURSES AND CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02)
- Microsoft Certified: Data Fundamentals (DP-900)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: AI Fundamentals (AI-900)
- Microsoft Certified: Azure Data Scientist Associate (DP-100)
- Oracle Agentic AI Foundations Associate
- SAP Certified: SAP Generative AI Developer
- NPTEL: Programming In Java
- NPTEL: Data Base Management System
- NPTEL: Cloud Computing
- NPTEL: Python for Data Science
- NPTEL: Data Science for Engineers

LANGUAGES

- Tamil (Native)
- English (Full Professional Proficiency)
- Hindi (Limited Working Proficiency)

EXTRACURRICULAR ACTIVITIES

- Finalist at HackIndia (Web3 and AI Hackathon)
- Technical Coordinator – AWS Cloud Clubs
- 3rd place at Hack-a-Cloud 3.0
- 1st place at AWS PartyRock Competition
- 2nd place at Ideathon Competition
- Technical Blogger on Medium

VOLUNTEER EXPERIENCE

Participated in an outreach program organized by college at children's and elderly welfare homes and led interactive sessions to support resident engagement:

- Anna Fathima Child Welfare Centre - Karapakkam
- Sri Sakthi Home - Kolathur
- Webbs Memorial - St. Thomas Mount